

Zhuqi Li

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EXPERIENCES

Member of Technical Staff @ xAI

11/2025 – Present

Inference for Computer Use Agent

- Designed and implemented computer-use agent model support in in-house SGLang, optimizing the inference engine to deliver a 24× speedup (1 to 24 FPS), enabling real-time inference on 720p video.

Senior Research Scientist @ ByteDance

02/2023 – 11/2025

xLLM Inference framework

- Implement vision language model (qwen2.5vl/Seed1.5-VL) support on xLLM, archiving 2%-15% better throughput while at the same latency compared with vLLM v1 on various GPUs. Adopted in production with 2000+ A100-equivalent compute.
- Implement Wan 2.1 model support on xLLM, serve with 14B model with TP+SP on 8xH20 GPUs
- Built post train infra support for [verl](#)+xLLM, including weight transfer/memory management. Enable and support post training with PPO and GRPO algorithms on DeepSeek-R1 671b and Qwen3 235b models on [verl](#)+xLLM.
- Built tool calling support for xLLM inference framework to support agentic tasks, supporting deployment on workloads consuming 500+ A100-equivalent compute.
- Developed the performance benchmarking as well as accuracy benchmarking test kit tests for xLLM. Automating regression detection and identifying 20+ critical performance/accuracy bugs pre-merge.

TikTok video processing system

- Led end-to-end design and implementation of Project CAE, TikTok's context adaptive video processing system. Play time/User +0.13%, Rebuffer time/Play time -4.7%, Ads value +0.437%. Paper accepted by [SIGCOMM 2025](#)
- Led and drive the design and implementation of TikTok's ROI-driven video transcoding scheduling system, reducing total transcoding CPU cores by 8.8% while maintaining the same user playback experience.

Ph.D. @ Princeton University

09/2017 – 01/2023

- Optimize QoE for short video streaming with user action modeling with Dashlet @ [NSDI 2023](#). Improve QoE by 28-101% and reduce the wasted video data by 30% compared with TikTok.
- Design REITS, a reflective tag to provide enhanced safety for self-driving vehicles. Improve FMCW radar sensing distance by 3.63x, Paper accepted by [HotMobile 2021](#), News coverage by [BBC](#), [New Scientist](#)
- Design reconfigurable network physical layer with LAIA @ [NSDI 2019](#). Achieve improvement on Shannon capacity by 51.4% and TCP throughput by 24%.

Research Intern @ Microsoft Research Redmond

06/2019 – 06/2020

- Build a video streaming and object detection pipeline on top of millimeter wave relay network for dense deployed wireless video surveillance system.
- Improve the recall by 41.5%-52.9% with same precision. Paper accepted by [ACM/IEEE SEC 2021](#); Patent number [11272423](#)

Software Engineer Intern @ Meta Platforms, Inc.

06/2021 – 08/2021

- Implement an MLP-based ML system with feature engineering and ensemble methods to predict wireless network quality.
- Reduce the MSE prediction error by 23.1% for signal strength and 26.7% for network throughput for the Facebook App.

EDUCATION

Ph.D. in Computer Science @ **Princeton University**

08/2017 – 03/2023

B.S. in Computer Science (Summa Cum Laude) @ **Peking University**

09/2013 – 07/2017

AWARDS AND HONORS

- GetMobile Research Highlights 2019 • Princeton Graduate Student Fellowship 2017 • Excellence Award of Internship @ Microsoft Research 2016 & 2017 • ICCSE Best Paper Award 2017 • Undergraduate Dissertation Award @ Peking University 2017 • Founder Scholarship, Peking University 2016

SKILLS

- Machine Learning:** PyTorch, Transformer, CNN, RNN, AutoML, Llama, Qwen, Deepseek
- LLM System:** xLLM, vLLM, SGLang, verl, Orca, Megatron-LM, FSDP
- Video Streaming:** Dash.js, H.264, H.265, H.266, ffmpeg, nginx
- System & Networking:** QUIC, TCP, Linux kernel, Mininet, DPDK
- Production:** A/B experiments, Grafana, CI/CD pipeline
- Programming Languages:** C/C++, Python, Java, SQL, JavaScript, Go, Android